MVP Architecture (High Level)

Overview

The MVP uses a modular client–server architecture.

- React frontend for maps, dashboards and forms
- Node.js/Express backend exposing a REST API
- PostgreSQL + PostGIS for relational and spatial data
- Deployed using Docker (Dev) and AWS (Prod)
- CI/CD with GitHub Actions

Users

Public Users · Contributors · Administrators

Frontend (React)

- Interactive Map
- Search & Filters
- Country & Company Profiles
- Analytics & Visualisations
- Contribute (Forms & Evidence)
- Admin Dashboard

HTTPS (REST/JSON)

Backend (Node.js / Express)

- Auth & Session
- Users & Roles
- Deployments Module
- Technologies Module
- Contributions & Verification
- Analytics Module
- Admin Module

Validation, Authorisation, Business Logic

Database

PostgreSQL + PostGIS

- Relational Data
- Spatial Data (PostGIS)
- Indexes & Relationships
- Future: Chunks (for RAG)

Future Extension (Not in MVP)

services/

Reserved for optional AI/RAG service

→ RAG / AI Service · Vector Store / Embeddings

Infrastructure & Tooling

Development

Docker Compose
Local development environment

CI/CD

GitHub Actions
Build, test, and continuous integration

Deployment (MVP)

Vercel
Frontend hosting

AWS (Future/Prod)
Backend API

PostgreSQL + PostGIS

Source Control & Project Management

GitHub
Repo, Issues, PRs

GitHub Projects
Planning & tracking